TQUK Functional Skills Qualification in Maths at Level 2

Examination Paper (Sample Assessment Paper 2)

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- Read each question **carefully**
- Answer **all** questions
- Write your answers **clearly** in the spaces provided
- **Check** your answers.

Information:

- This examination has **two** sections. These are clearly labelled
- You are **not allowed** to use a calculator for Section A
- You **are allowed** to use a calculator for Section B
- The **maximum** mark for this examination is 60
- The marks available for each question are shown in **bold** beneath each question.

SAMPLE

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

- Assessment: Sample Paper 2 (Non-Calculator & Calculator)
- Complete stepped calculations, methods & mark scheme codes [M1/A1]
- Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)
- Pass standard: 36 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

Section B	45		
Total marks	60		

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Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 0.872 + 0.15

Step 1: Align decimal places with padding zero:

0.872 + 0.150 = 1.022

Answer 1.022 ✓ *[1 mark]*

[1]

2. Work out 2.205 × 4

Step 1: Long multiplication:

2.205 × 4 = 8.820 = 8.82

Answer 8.82 ✓ *[1 mark]*

[1]

SAMPLE

Questions continue on the following page

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3. A cyclist cycles 3 miles in 10 minutes.

What is the cyclist's average speed in miles per hour (mph)?

Step 1: Convert time to hours or scale factor (60 mins = 1 hour):

Scale factor = 60 mins ÷ 10 mins = 6

Step 2: Calculate speed:

Speed = 3 miles × 6 = 18 mph (or 3 ÷ (10/60) = 18)

Answer 18 ✓ *[2 marks]*	**mph**

[2]

4. Write 0.86:

(a) as a fraction

(b) as a percentage.

Step 1: Write as fraction: 0.86 = 86/100 = 43/50

Step 2: Convert to percentage: 0.86 × 100% = 86%

[2 marks]

Answer	**(a) Fraction:** 86/100 ✓	**(b) Percentage:** 86% ✓

[2]

SAMPLE

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5. Jordan goes out for lunch.

The bill comes to £20

Jordan wants to leave extra money as a tip.

The extra money is 18% more than the bill.

What is the total amount that Jordan will pay?

Step 1: Find 18% of £20:

10% of £20 = £2.00 | 1% = £0.20 --> 8% = £1.60

Tip = £2.00 + £1.60 = £3.60 (or £20 × 0.18 = £3.60)

Step 2: Total amount paid = Bill + Tip:

£20.00 + £3.60 = £23.60

Answer	£23.60 ✓ *[2 marks]*

[2]

SAMPLE

Questions continue on the following page

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6. Work out the median of these numbers:

5000 3000 2500 4250 5500 6600

Step 1: Order numbers from lowest to highest:

2500, 3000, 4250, 5000, 5500, 6600

Step 2: Middle two numbers are 4250 and 5000 (even count n = 6):

Median = (4250 + 5000) ÷ 2 = 9250 ÷ 2 = 4625

Answer	4625 ✓ *[2 marks]*

[2]

SAMPLE

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7. A map has a scale of 1 : 50 000

A school and a supermarket are 2.5 centimetres (cm) apart on the map.

How far apart are the school and supermarket in real life?

Give the units in your answer.

Step 1: Calculate real-life distance in cm:

2.5 cm × 50 000 = 125 000 cm

Step 2: Convert to metres (÷ 100):

125 000 cm ÷ 100 = 1250 m

Step 3: Convert to kilometres (÷ 1000):

1250 m ÷ 1000 = 1.25 km

Answer	1.25 km (or 1250 m) [2 marks]

[2]

SAMPLE

Questions continue on the following page

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8. A company gives its staff 6 pizzas to share.

Each pizza is cut into 8 slices.

On the first day, the staff eat 3 whole pizzas and 5 additional slices.

On the next day, they eat another 10 slices of pizza.

How much pizza is left after the second day?

Give your answer as a mixed number.

Step 1: Total pizza slices available:

6 pizzas × 8 slices = 48 slices

Step 2: Slices eaten across both days:

Day 1: (3 × 8) + 5 = 29 slices | Day 2: 10 slices

Total eaten = 29 + 10 = 39 slices (or 3 5/8 + 1 2/8 = 4 7/8)

Step 3: Slices left as a mixed number:

48 - 39 = 9 slices left --> 9/8 = 1 1/8 pizzas

Answer	1 1/8 ✓ [3 marks]

[3]

SAMPLE

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End of Section A.

SAMPLE

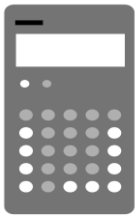
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Section B begins on Page 8.

SAMPLE

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Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. Put these numbers in order starting with the lowest:

–865 –811 –820 –814

Show any working here:

Step 1: On a negative number line, values further below 0 are smaller:

-865 < -820 < -814 < -811

Answer	*-865* , *-820* ,	*-814* , *-811*	✓	*[1 mark]*
	Lowest	→		**Highest**

[1]

2. There are 2.54 centimetres (cm) in 1 inch.

How many inches are there in 127 cm?

Step 1: Divide by conversion factor:

127 cm ÷ 2.54 cm/inch = 50 inches

Answer	*50* ✓	*[1 mark]*

[1]

Questions continue on the following page

SAMPLE

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3. Calculate $12^2 - 3 \times 2.1$

Step 1: Evaluate index: 12 squared = 12 × 12 = 144

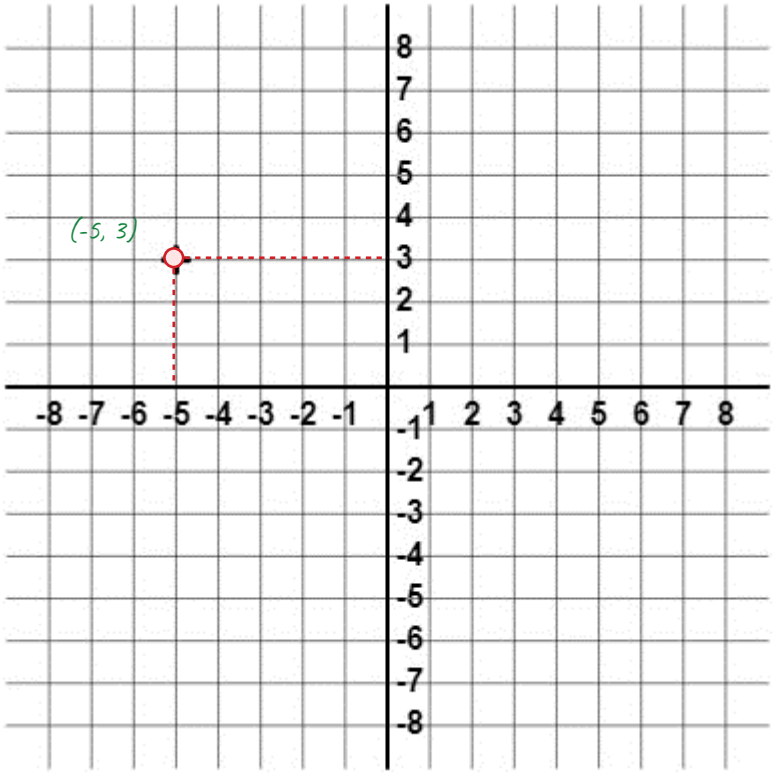
Step 2: Multiplication: 3 × 2.1 = 6.3

Step 3: Subtraction: 144 – 6.3 = 137.7

Answer	137.7 ✓ [1 mark]

[1]

4. Write down the coordinates of the point labelled on the grid.



Answer	✓ (-5 , 3)

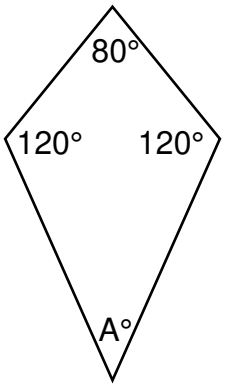
[1 mark]

[1]

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5. Work out the value of angle A.

Not drawn to scale



Step 1: Interior angles of any quadrilateral sum to 360 degrees:

Sum of known angles = 120° + 120° + 80° = 320°

Step 2: Calculate missing angle A:

Angle A = 360° - 320° = 40°

Answer 40 ✓ *[2 marks]* °

[2]

SAMPLE

Questions continue on the following page

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6. A garden has a circular flowerbed in the centre.

The diameter of the flowerbed is 1.38 metres (m).

Children make fairy lights to go all around the edge of the flowerbed.

The total length of the fairy lights is 4.5 metres (m).

Are the fairy lights long enough to decorate all around the edge of the flowerbed?

Show how you decide.

Use $\pi = 3.14$

Step 1: Circumference formula C = pi × d (using pi = 3.14):

C = 3.14 × 1.38 m = 4.3332 m

Step 2: Compare circumference with fairy lights length (4.5 m):

4.3332 m < 4.5 m (extra length = 4.5 – 4.3332 = 0.1668 m)

Fairy lights are longer than the perimeter. Therefore Yes.

Answer	Yes (4.33 m < 4.5 m) [3 marks]

[3]

SAMPLE

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7. Charlie recently used 14 litres of paint to decorate the living room walls.

The total area painted was 91 m^2

Charlie now wants to paint the bedroom walls.

The total area to be painted is 78 m^2

Charlie has 11 litres of paint.

Does Charlie have enough paint?

Show how you decide.

Step 1: Find paint coverage rate per litre from living room:

Coverage rate = 91 m² ÷ 14 litres = 6.5 m² per litre

Step 2: Calculate area covered by 11 litres:

Area = 11 litres × 6.5 m²/litre = 71.5 m²

Step 3: Compare with bedroom area to be painted (78 m^2):

71.5 m² < 78 m² (shortfall of 6.5 m²; needs 78 ÷ 6.5 = 12 litres)

Charlie does not have enough paint: No.

Answer	No (71.5 m² ✓) *[3 marks]*

[3]

SAMPLE

Questions continue on the following page

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8. Morgan is planning a holiday and finds this information:

Greenland	Average temperature 14° Fahrenheit (°F)
Iceland	Average temperature –11° Celsius (°C)

Morgan wants to go to the warmer country.

Which country should Morgan go to?

Show how you decide.

Use: $C = \frac{5(F - 32)}{9}$

Where: C = temperature in degrees Celsius
F = temperature in degree Fahrenheit

Step 1: Convert Greenland temperature (F = 14°F) to Celsius:

$C = 5 \times (14 - 32) \div 9 = 5 \times (-18) \div 9 = -90 \div 9 = -10°C$

Step 2: Compare temperatures (Morgan wants the warmer country):

$-10°C > -11°C$ (Greenland is warmer by 1°C). Morgan should visit Greenland.

Answer	Greenland (-10° ✓ -11°C) [3 marks]

[3]

SAMPLE

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9. This year, Jo's gym membership price has increased by 5%.

The new monthly price is £33.60

How much did Jo pay for an entire year's membership last year?

Step 1: Original monthly price last year (reverse percentage):

105% = £33.60 --> 100% = £33.60 ÷ 1.05 = £32.00 per month

Step 2: Full year's membership cost (12 months):

Annual Cost = £32.00 × 12 = £384.00

Answer	£ 384 ✓ *[3 marks]*

[3]

SAMPLE

Questions continue on the following page

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10. An athletics club awards points each year to its members.

In 2021, the club awarded its members a mean of 90.3 points in the year.

The grouped frequency table shows the points the club awarded its members in 2022:

Number of points awarded	Frequency
0 ≤ points < 40	7
40 ≤ points < 80	34
80 ≤ points < 120	100
120 ≤ points < 160	9
Total	**150**

The club president says:

'The mean number of points per member in 2022 has increased since 2021'

Is the president correct?

Show how you decide.

Step 1: Midpoints of groups: 20, 60, 100, 140

Step 2: Calculate sum of (frequency × midpoint):

0–40: 7 × 20 = 140 | 40–80: 34 × 60 = 2040

80–120: 100 × 100 = 10 000 | 120–160: 9 × 140 = 1260

Total points = 140 + 2040 + 10 000 + 1260 = 13 440

Step 3: Estimated mean per member in 2022:

Mean = 13 440 ÷ 150 = 89.6 points

Step 4: Compare with 2021 mean (90.3 points):

89.6 < 90.3 (points decreased, did not increase). President is incorrect: No.

Answer	No (Mean = ✓ < 90.3) [4 marks]

[4]

SAMPLE

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11. Hayden is organising activities for 80 students going on a school trip.

The ratio of the number of students who can participate in swimming, rock climbing and go karting is 5 : 3 : 2

Complete the activity list for Hayden.

Activity	Number of Students
Swimming	40
Rock climbing	24
Go karting	16
Total	80

Show any workings here:

Step 1: Total ratio parts = 5 + 3 + 2 = 10 parts

Step 2: Value per part = 80 students ÷ 10 parts = 8 students per part

Step 3: Calculate number of students for each activity:

Swimming = 5 parts × 8 = 40 students

Rock climbing = 3 parts × 8 = 24 students

Go karting = 2 parts × 8 = 16 students

Check total: 40 + 24 + 16 = 80 students

[4 marks]

[4]

SAMPLE

Questions continue on the following page

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12. Harper is a manager of a company.

Last year the company made a profit of nine hundred and seventy-five thousand and fifty pounds.

This year Harper wants to increase the profit by 3.5%.

Harper thinks the new profit will be more than £1 010 000

Is Harper correct?

Show how you decide.

Step 1: Write profit in figures:

Nine hundred and seventy-five thousand and fifty = £975 050

Step 2: Calculate 3.5% profit increase:

Increase = £975 050 × 0.035 = £34 126.75

New Profit = £975 050 + £34 126.75 = £1 009 176.75

(or £975 050 × 1.035 = £1 009 176.75)

Step 3: Compare with target (£1 010 000):

£1 009 176.75 < £1 010 000 (target missed by £823.25)

Therefore Harper is not correct: No.

Answer	No (£1 009 176.75 < £1 010 000) *[4 marks]*

[4]

SAMPLE

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13. A delivery driver keeps a record of the number of miles per gallon (mpg) their car achieves each day.

These are the results for week 2:

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
30	38	22	37	40	32	39

This is a summary of the results for week 1:

Week 1	
Mean mpg	32
Range	16

The delivery driver claims:

a) *'on average, the car achieved more miles per gallon in week 2 than in week 1'*
b) *'the miles per gallon achieved in week 2 were more consistent than in week 1.'*

Are **each** of the driver's claims correct?

Give reasons for your answers.

Show your working.

Show any workings here:

Step 1: Week 2 Mean = (30+38+22+37+40+32+39) ÷ 7 = 238 ÷ 7 = 34 mpg

Step 2: Week 2 Range = 40 – 22 = 18 mpg (vs Week 1 Range = 16 mpg)

Mean: 34 > 32 (higher average) | Range: 18 > 16 (larger spread, less consistent)

Comment regarding claim a)

Yes, claim a) is correct: Week 2 mean (34 mpg) is higher than Week 1 mean (32 mpg).

On average, the delivery driver achieved 2 more miles per gallon in week 2.

Comment regarding claim b)

No, claim b) is incorrect: Week 2 range (18 mpg) is greater than Week 1 range (16 mpg).

A larger range indicates greater variation, so fuel consumption was less consistent.

[4 marks]

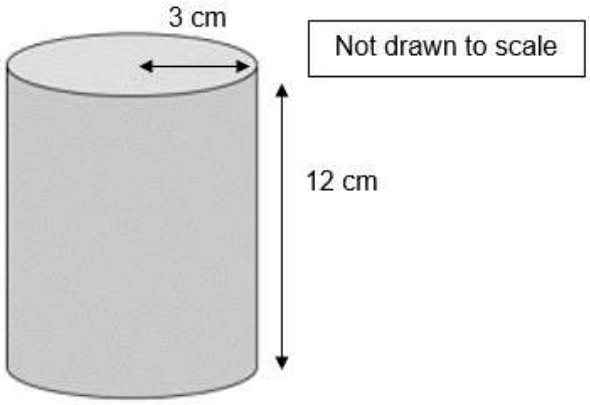
[4]

Questions continue on the following page

SAMPLE

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14. A nursery asks Cameron to make 50 cylinder-shaped toy blocks using this design:



When the blocks are made, Cameron needs to paint them using glow-in-the-dark paint.

Each tin has enough paint to cover 1150 square centimetres (cm^2).

How many tins of paint will Cameron need to buy to paint all 50 toy blocks?

Use $\pi = 3.14$

Step 1: Surface Area of 1 cylinder block (radius r = 3 cm, height h = 12 cm):

Two circular bases = 2 × (pi × r²) = 2 × (3.14 × 3²) = 2 × 28.26 = 56.52 cm²

Curved surface area = 2 × pi × r × h = 2 × 3.14 × 3 × 12 = 226.08 cm²

Total Surface Area per block = 56.52 + 226.08 = 282.6 cm²

Step 2: Total area for all 50 toy blocks:

Total Area = 50 blocks × 282.6 cm² = 14 130 cm²

Step 3: Number of tins needed (each tin covers 1150 cm²):

Tins = 14 130 ÷ 1150 = 12.2869... tins

Step 4: Must buy whole tins (round up to ensure full coverage):

12.28... rounds up to 13 tins

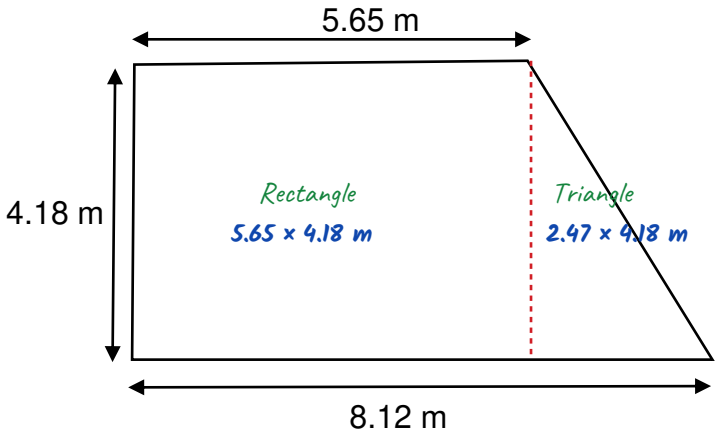
Answer 13 *[5 marks]* **tins**

[5]

SAMPLE

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15. A family wants to cover this garden with grass.



Not drawn to scale

They will buy grass as turf in square metres (m^2).

The family finds these prices online:

Cost per square metre (m^2) Must buy a whole number of square metres					
£5.00	£3.80	£4.50	£5.00	£4.25	£6.00

The family wants to pay the same amount as the modal cost.

How much will it cost to cover the entire garden in turf?

Show your working.

Step 1: Calculate garden area (Rectangle + Right-angled Triangle):

Triangle base = 8.12 m − 5.65 m = 2.47 m

Triangle Area = 0.5 × 2.47 × 4.18 = 5.1623 m²

Rectangle Area = 5.65 × 4.18 = 23.617 m²

Total Garden Area = 5.1623 + 23.617 = 28.7793 m²

Step 2: Whole number of square metres required:

28.7793 m² rounds up to 29 m²

Step 3: Modal price from price list (£5.00, £3.80, £4.50, £5.00, £4.25, £6.00):

£5.00 occurs most frequently (frequency = 2) --> Mode = £5.00 per m²

Step 4: Total turf cost = 29 m^2 × £5.00 = £145.00

Answer £ 145 ✓ *[6 marks]*

[6]

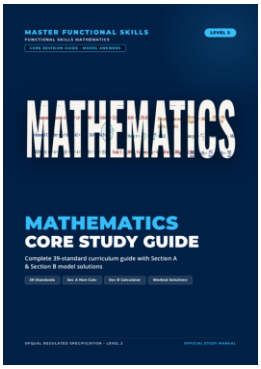
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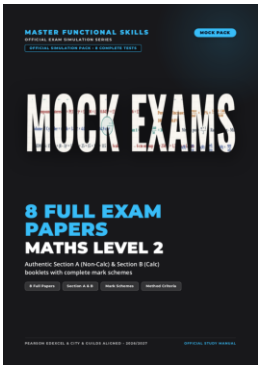


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